

Receptionist Module

Frontend Developer Guide

Version: 1.0 | Date: July 2026 | Audience: Frontend developers
Backend: HM-System (FastAPI) | Base URL: /receptionist | Auth: JWT Bearer

1. Purpose & Scope

The Receptionist module manages the live OPD waiting line after OPD Billing has booked an appointment. Reception checks in arriving patients, monitors doctor queues, answers doctor 'next patient' requests, and handles no-show / rejoin actions.

This document is written for frontend developers building the receptionist UI in HM-frontend-Side. All 11 backend REST endpoints are complete; frontend screens are not yet implemented.

What reception DOES

- Show today's scheduled arrivals (not yet checked in)
- Check-in patients into the doctor queue (assigns token number)
- View today's queue across all doctors or per-doctor boards
- Answer doctor 'next patient' requests (call patient to room)
- Mark no-show and rejoin patients
- Queue history reporting and CSV export

What reception does NOT do

- Register patients or collect payment (OPD Billing module)
- Create appointments (OPD Billing module)
- Start or complete consultations (Doctor module)
- Record vitals (Nurse module - optional pre-consult step)

2. End-to-End Workflow

```
OPD Billing -> POST /opd/appointments -> appointment (scheduled)
Patient arrives at hospital
Reception -> POST /receptionist/check-in/{appointment_id}
              -> patient_queue row (status=waiting) + token number
[Optional] Nurse -> POST /nurse/vitals -> queue status vitals_completed
Doctor -> PUT /queue/complete/{queue_id} (finish current patient)
Doctor -> POST /queue/request-next -> pending call request
Reception -> GET /receptionist/pending-calls (poll every 5-10s)
Reception -> POST /receptionist/call-patient/{queue_id}
              -> status=called, called_at, called_by set
Doctor -> PUT /queue/start/{queue_id} -> in_progress
Doctor -> PUT /queue/complete/{queue_id} -> completed
Repeat...
```

Appointment status flow

```
scheduled -> waiting -> in_progress -> completed
           |           |
           |           |
        check-in   doctor start
```

waiting -> cancelled (no-show path)

Queue status flow

(none) -> waiting -> [vitals_completed] -> called -> in_progress -> completed

waiting | vitals_completed | called -> no_show -> rejoin -> waiting (new token at end)

Key business rules

- Booking an appointment does NOT put the patient in the queue - check-in is separate.
- Reception cannot call a patient unless the doctor first POST /queue/request-next.
- FIFO by token number; urgent/emergency priority sorts first.
- Only one in_progress patient per doctor at a time.
- Duplicate check-in returns HTTP 409.
- All dates/times use IST (Asia/Kolkata). API returns ISO 8601 with timezone.
- No WebSocket - pending calls use HTTP polling (5-10 second interval).

3. Authentication & Permissions

Auth mechanism

- Header: Authorization: Bearer <JWT>
- 401 - invalid or missing token -> redirect to /login
- 403 - Permission denied: {permission} required

Receptionist role permissions (from seed.py)

- patients:view
- opd:view
- appointments:view
- appointments:update

Permission mapping per endpoint

Permission	Endpoints
opd:view	dashboard, today-queue, doctor-queue,
	pending-calls, queue-history, export
appointments:view	arrivals
appointments:update	check-in, call-patient, no-show, rejoin

Frontend role constant: RECEPTIONIST = 'receptionist' (src/shared/constants/index.js). Route label: 'Reception'. Folder: src/pages/receptionist/ (to be created).

4. API Endpoints Summary

#	Method	Path	Permission
1	GET	/receptionist/dashboard	opd:view
2	GET	/receptionist/today-queue	opd:view
3	GET	/receptionist/arrivals	appointments:view
4	POST	/receptionist/check-in/{appointment_id}	appointments:update
5	GET	/receptionist/doctor-queue/{doctor_id}	opd:view
6	GET	/receptionist/pending-calls	opd:view
7	POST	/receptionist/call-patient/{queue_id}	appointments:update
8	PATCH	/receptionist/queue/{queue_id}/no-show	appointments:update
9	PATCH	/receptionist/queue/{queue_id}/rejoin	appointments:update
10	GET	/receptionist/queue-history	opd:view
11	GET	/receptionist/queue-history/export	opd:view

Swagger docs available at /docs under tag 'Receptionist'.

5. API Details - Read Endpoints

GET /receptionist/dashboard

- Query: doctor_id (optional int) - filter stats to one doctor

```
{
  "success": true,
```

Resp

```
"data": {
  "total_patients": 45,
  "waiting": 10, "called": 2, "in_progress": 3,
  "completed": 30, "no_show": 2,
  "pending_doctor_requests": 2,
  "todays_arrivals": 52, "todays_checked_in": 45,
  "todays_cancelled": 3,
  "average_waiting_time_minutes": 18.5
}
}
average_waiting_time_minutes is null if no consultations started today.
```

GET /receptionist/today-queue

- All patients checked in today across all doctors.
- Query: doctor_id, doctor_name (partial), patient_id, status, search, page (def 1), limit (def 20, max 100)
- Search matches: patient name, UHID, phone, token, appointment UID, doctor name
- Sort: priority DESC -> called/waiting/vitals_completed first -> token ASC

```
{
  "success": true, "queue_date": "2026-06-23",
  "total": 45, "page": 1, "limit": 20,
  "queue": [ { QueueItemOut + doctor_name } ]
}
```

GET /receptionist/arrivals

- Today's appointments with status=scheduled and NO queue row yet.
- Query: doctor_id, search, page (def 1), limit (def 20, max 100)
- Search matches: patient name, UHID, phone, appointment UID

```
{
  "success": true, "total": 45, "page": 1, "limit": 20,
  "arrivals": [{
    "appointment_id": 42, "appointment_uid": "AP-0042",
    "patient_id": 10, "patient_name": "Nilesh Patil",
    "patient_uid": "P-1001", "patient_phone": "9876543210",
    "doctor_id": 5, "doctor_name": "Dr. Sharma",
    "scheduled_at": "2026-06-23T10:30:00+05:30"
  }]
}
```

GET /receptionist/doctor-queue/{doctor_id}

- Query: status, search, date (default today IST), page, limit
- If page and limit omitted, returns all rows for that doctor/date

```
{
  "success": true, "doctor_id": 5,
  "total": 3, "page": 1, "limit": 3,
  "queue": [ { QueueItemOut } ]
}
```

GET /receptionist/pending-calls

- Query: doctor_id (optional)
- Poll every 5-10 seconds while screen is open and tab is focused

```
{
  "success": true, "total": 1,
  "pending_calls": [{
    "request_id": 3, "doctor_id": 5, "doctor_name": "Dr. Sharma",
    "queue_id": 16, "queue_number": 7,
    "appointment_id": 42, "patient_id": 10,
    "patient_name": "Nilesh Patil", "patient_uid": "P-1001",
    "appointment_time": "10:30:00", "status": "pending",
    "requested_at": "2026-06-23T10:30:00+05:30"
  }]
}
```

GET /receptionist/queue-history

- Query: date OR date_from/date_to, doctor_id, status, search, page, limit
- Default date range: today only

```
{
  "success": true,
  "date_from": "2026-06-01", "date_to": "2026-06-23",
  "total": 120, "page": 1, "limit": 20,
  "history": [ { QueueHistoryItem = QueueItemOut + doctor_name } ]
}
```

GET /receptionist/queue-history/export

- Same query params as history + format=csv (only csv supported)
- Returns raw CSV file download (Content-Disposition: attachment)
- Columns: Queue Date, Token, Appointment UID, Patient, UHID, Phone,
- Doctor, Status, Checked In, Called At, Called By (ID/Name),
- Consultation Started, Consultation Completed

6. API Details - Write Endpoints**POST /receptionist/check-in/{appointment_id}**

- Request body: none
- Creates patient_queue row (status=waiting, assigns token_number)
- Sets appointments.status = waiting

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- 201 on success; 404 if not found; 409 if already checked in today

```
{
  "success": true,
  "message": "Patient checked in successfully",
  "queue": { QueueItemOut }
}
```

POST /receptionist/call-patient/{queue_id}

- Request body: none
- Requires pending doctor request (doctor must POST /queue/request-next first)
- Sets status=called, called_at, called_by; fulfills matching next-request
- 400 if no pending request, wrong status, or doctor already has in_progress patient

```
{
  "success": true,
  "message": "Patient called to doctor room",
  "queue": {
    "queue_id": 16, "queue_number": 7, "status": "called",
    "called_at": "2026-06-23T10:32:00+05:30",
    "called_by": 12, "called_by_name": "Priya Reception"
  }
}
```

PATCH /receptionist/queue/{queue_id}/no-show

- Allowed when status is waiting, vitals_completed, or called
- Sets status=no_show; cancels appointment; cancels pending next-request

PATCH /receptionist/queue/{queue_id}/rejoin

- Allowed only when status is no_show
- Sets status=waiting; assigns new token at end of line; appointment -> waiting

7. Shared Data Types

QueueItemOut (used in most queue responses)

```
{
  queue_id: number           // DB patient_queue.id
  appointment_id: number
  appointment_uid?: string
  queue_number: number       // token_number in DB
  patient_id: number
  patient_name: string
  patient_uid: string        // UHID
  patient_phone?: string
  doctor_id: number
  status: QueueStatus        // lowercase string
  checked_in_at?: datetime   // queue_entered_at in DB
  called_at?: datetime
  called_by?: number         // receptionist user id
  called_by_name?: string
  consultation_started_at?: datetime
  consultation_completed_at?: datetime
  queue_date?: date
  doctor_name?: string       // today-queue & history only
}
```

QueueStatus enum values

Value	Meaning	Badge color suggestion
waiting	Checked in, in waiting area	Blue
vitals_completed	Nurse recorded vitals	Teal
called	Reception called to doctor room	Orange
in_progress	Doctor started consultation	Purple
completed	Consultation finished	Green
cancelled	Cancelled	Gray
no_show	Patient absent	Red

AppointmentStatus (appointments table)

scheduled | waiting | in_progress | completed | cancelled

NextRequestStatus (doctor_queue_next_requests)

pending | fulfilled | cancelled

QueuePriority

normal | urgent | emergency (sorts higher priority first)

8. Frontend Screens to Build

#	Screen	Route	Primary API
1	Dashboard	/receptionist/dashboard	GET /dashboard
2	Arrivals	/receptionist/arrivals	GET /arrivals + POST check-in

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2b	Today's Queue	/receptionist/today	GET /today-queue
3	Doctor Queues	/receptionist/queues	GET /doctor-queue/{id}
4	Pending Calls	/receptionist/pending-calls	GET pending + POST call
5	Queue History	/receptionist/history	GET history + export

Sidebar menu

- Dashboard
- Arrivals
- Today's Queue
- Doctor Queues
- Pending Calls (badge when pending_doctor_requests > 0)
- Queue History
- Sign out

9. Screen Specifications

Screen 1 - Dashboard (/receptionist/dashboard)

- Stat cards: waiting, called, in_progress, completed, no_show, pending_doctor_requests
- Manager metrics: todays_arrivals, todays_checked_in, todays_cancelled, avg waiting time
- Optional doctor filter (?doctor_id=5)
- Quick action links to Arrivals, Today's Queue, Doctor Queues, Pending Calls
- Optional auto-refresh every 30-60 seconds

Screen 2 - Arrivals (/receptionist/arrivals)

- Table: Time, Appointment UID, Patient, UHID, Phone, Doctor, Check-in button
- Filters: search (debounce 400ms), doctor dropdown, pagination (page/limit)
- Doctor dropdown data: GET /opd/departments + GET /opd/doctors/department/{id}
- On check-in: POST /check-in/{appointment_id} -> toast 'Token #N checked in' -> remove row
- Handle 409: 'Patient already checked in today'
- Empty state: 'No arrivals waiting for check-in today'

Screen 2b - Today's Queue (/receptionist/today)

- All checked-in patients across all doctors today
- Filters: doctor_id, doctor_name, patient_id, status tabs, search, pagination
- Columns: Token, Appointment UID, Patient, UHID, Doctor, Status, Checked in, Called, Called by

Screen 3 - Doctor Queues (/receptionist/queues)

- Requires doctor selection before loading
- Filters: status tabs, search, date picker (default today)
- Row actions: No-show (waiting/called/vitals_completed), Rejoin (no_show only)
- Do NOT add 'Call patient' here - use Pending Calls screen
- Empty: 'Select a doctor' or 'No patients in queue for Dr. {name} today'

Screen 4 - Pending Calls (/receptionist/pending-calls)

- Poll GET /pending-calls every 5-10 seconds (tab focused)
- Card per request: doctor, patient, UHID, token, requested_at, Call patient button
- On call: POST /call-patient/{queue_id} -> status becomes 'called'

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- Sidebar badge when total > 0; optional sound on new request_id
- Empty: 'No doctors waiting for patient call'

Screen 5 - Queue History (/receptionist/history)

- Date range filters, doctor, status, search, pagination
- Export button: GET /queue-history/export?format=csv with same filters
- For today live data, use Today's Queue or Doctor Queues instead

10. Global UI Conventions

Pattern	Rule
Auth	Redirect to /login on 401
Permissions	Hide module if role != receptionist
Search debounce	400ms
Default page size	20 (max 100)
Pending calls poll	5-10s while tab focused
Dashboard refresh	Optional 30-60s
Date/time display	IST; API returns ISO 8601 with timezone
Error toasts	Show API detail message
Status badges	Consistent colors across all screens

Filters, search & pagination summary

Screen	Search	Filters	Pagination	Polling
Dashboard	No	Doctor	No	Optional
Arrivals	Yes	Doctor	Yes	No
Today's Queue	Yes	Doctor, status	Yes	Manual
Doctor Queues	Optional	Doctor, status, date	Optional	Manual
Pending Calls	No	Doctor	No	Yes 5-10s
Queue History	Yes	Date, doctor, status	Yes	Export CSV

11. TypeScript Types (src/types/receptionist.ts)

```

export type PaginatedMeta = { total: number; page: number; limit: number }

export type ReceptionistDashboard = {
  total_patients: number; waiting: number; called: number
  in_progress: number; completed: number; no_show: number
  pending_doctor_requests: number; todays_arrivals: number
  todays_checked_in: number; todays_cancelled: number
  average_waiting_time_minutes: number | null
}

export type Arrival = {
  appointment_id: number; appointment_uid: string
  patient_id: number; patient_name: string; patient_uid: string
  patient_phone?: string; doctor_id: number; doctor_name: string
  scheduled_at: string
}

export type QueueItem = {
  queue_id: number; appointment_id: number; appointment_uid?: string
  queue_number: number; patient_id: number; patient_name: string
  patient_uid: string; patient_phone?: string; doctor_id: number
  status: string; checked_in_at?: string; called_at?: string | null
  called_by?: number | null; called_by_name?: string | null
  consultation_started_at?: string | null
  consultation_completed_at?: string | null; doctor_name?: string
}

```

```
export type PendingCall = {
  request_id: number; doctor_id: number; doctor_name: string
  queue_id: number; queue_number: number; patient_name: string
  patient_uid: string; requested_at: string
}
```

12. API Client (src/api/receptionist.ts)

```
getDashboard(params?: { doctor_id?: number })
getTodayQueue(params?: {
  doctor_id?: number; doctor_name?: string; patient_id?: number
  status?: string; search?: string; page?: number; limit?: number
})
getArrivals(params: { doctor_id?: number; search?: string; page?: number; limit?: number })
checkIn(appointmentId: number)
getDoctorQueue(doctorId: number, params?: { status?: string; search?: string; date?: string })
getPendingCalls(params?: { doctor_id?: number })
callPatient(queueId: number)
markNoShow(queueId: number)
rejoinQueue(queueId: number)
getQueueHistory(params: {
  date?: string; date_from?: string; date_to?: string
  doctor_id?: number; status?: string; search?: string
  page?: number; limit?: number
})
exportQueueHistory(params: { ...same filters...; format?: 'csv' })
```

13. Suggested File Structure

```
pages/receptionist/
  ReceptionistRoutes.tsx
  Dashboard.tsx
  Arrivals.tsx
  TodayQueue.tsx
  DoctorQueues.tsx
  PendingCalls.tsx
  QueueHistory.tsx
hooks/
  useReceptionistArrivals.ts
  useDoctorQueue.ts
  usePendingCallsPoll.ts
components/
  QueueTable.tsx
  ArrivalTable.tsx
  ArrivalFilters.tsx
  DoctorQueueFilters.tsx
  PendingCallCard.tsx
  QueueStatusBadge.tsx
  PaginationBar.tsx
```

14. Error Handling

HTTP	When	Frontend action
400	Invalid queue state, no pending request	Toast with detail
401	Invalid/missing token	Redirect to /login
403	Missing permission	Toast: contact admin
404	Appointment/queue not found	Toast with detail
409	Duplicate check-in	Toast: already checked in

Common API detail messages

- Patient already checked in for this appointment today
- Patient already has a queue entry with this doctor today
- No pending doctor request for this patient...
- No-show is only allowed for patients waiting or called...
- Only no-show patients can rejoin the queue

15. Integration with Other Modules

OPD Billing (/opd) - upstream

- POST /opd/appointments - creates scheduled appointments (before check-in)
- POST /opd/patient/register - patient registration (not reception)
- GET /opd/departments - department dropdown
- GET /opd/doctors/department/{department_id} - doctor dropdown
- DO NOT USE: GET /opd/visits/today or GET /opd/queue/today (billing visits, not clinical queue)

Doctor module (/queue) - downstream partner

- POST /queue/request-next - doctor signals ready; creates pending call
- PUT /queue/start/{queue_id} - called -> in_progress
- PUT /queue/complete/{queue_id} - finishes consultation
- GET /queue/current - doctor's active patient

Nurse module - optional

- POST /nurse/vitals - updates queue waiting -> vitals_completed

Deprecated endpoints - do NOT use in new frontend

Old endpoint	Use instead
POST /queue/add	POST /receptionist/check-in/{appointment}
GET /opd/queue/next-requests	GET /receptionist/pending-calls
POST /opd/queue/send-in	POST /receptionist/call-patient/{queue_id}

16. Call Patient vs Start Consultation

Action	Actor	Sets	Meaning
Call patient	Reception	called, called_at, called_by	Patient sent to room
Start consultation	Doctor	in_progress, consultation_started_at	Clinical encounter begins
Complete	Doctor	completed, consultation_completed_at	Visit finished

17. Recommended Build Order

Phase	Screens	Why
P1	Arrivals + Check-in	Core workflow entry point
P2	Doctor Queues + Pending Calls	Live queue + polling
P3	Dashboard	Summary after core flows work
P4	Queue History	Reporting
P5	Polish	Badges, sounds, empty states

18. Backend Source Files (Reference)

- Routers/receptionist_router.py - 11 HTTP endpoints
- Services/receptionist_service.py - business logic orchestration
- Schemas/receptionist_schema.py - Pydantic request/response models
- Models/doctor_patient_queue.py - PatientQueue, QueueStatus, QueuePriority
- Models/doctor_queue_next_request.py - DoctorQueueNextRequest
- Models/opd_billing.py - Appointment, AppointmentStatus
- main.py - app.include_router(receptionist_router)

Related markdown docs: [HM-System/Docs/frontend/roles/receptionist.md](#), [HM-System/Docs/flows/receptionist-module.md](#), [HM-System/Docs/frontend/API-REFERENCE.md](#) (Section 4).